Lab Assignment 9
Web Technology Lab - Blog website using MERN stack

Kalpana Bhaskar - AI DS:A (24011101049)

Aim:

To develop a full-stack Blog/Post Management web application using React, Express.js, Node.js, and MongoDB Atlas, implementing CRUD operations over a REST API using the native MongoDB Node.js driver and the browser's native fetch() method.

Project Structure:

```
project/
├── app/
│   └── src/
│       ├── App.jsx
│       ├── components/
│       │   └── PostSummary.jsx
│       └── pages/
│           ├── Home.jsx
│           ├── Create.jsx
│           ├── Post.jsx
│           └── Archive.jsx
│
└── server/
    ├── .env
    ├── db/
    │   └── conn.mjs
    ├── index.mjs
    ├── loadEnvironment.mjs
    └── routes/
        └── posts.mjs
```

Design / Algorithm:

1. Configure environment variables in server/.env for the MongoDB Atlas connection string and server port.

2. Load environment variables using loadEnvironment.mjs and establish a persistent connection to MongoDB Atlas in server/db/conn.mjs using the native MongoDB Node.js driver (MongoClient).

3. Initialize the Express server in server/index.mjs, mounting CORS, JSON parsing middleware, and the posts router (routes/posts.mjs).

4. Define RESTful API routes in routes/posts.mjs handling GET / (fetch all posts), GET /:id (fetch single post), POST / (create post), PATCH /:id or PUT /:id (update post), and DELETE /:id (delete post).

5. Set up React client routing in App.jsx to manage navigation between Home, Create, Post, and Archive pages.

6. In Home.jsx and Archive.jsx, use the native fetch() API inside useEffect hooks to retrieve post listings and render them using the PostSummary component.

7. In Create.jsx, capture title and content input, then issue a POST request using fetch() to insert a new document into MongoDB Atlas.

8. In Post.jsx, query a single post by ID using fetch(), displaying full post details with options to update or delete the record.

9. Dynamically re-render UI state upon successful HTTP response status codes.

Frontend Components & Pages (React):

- App.jsx: Main application container handling route definitions and navigation layout.
- PostSummary.jsx: Reusable UI component for displaying individual post cards/summaries in listings.
- Home.jsx: Main page displaying active post listings fetched from the database.
- Create.jsx: Page containing form inputs for creating and submitting new blog posts.
- Post.jsx: Detailed view page for displaying, editing, and deleting an individual blog post.
- Archive.jsx: View page dedicated to archived or historical blog posts.

Frontend Functionality (React):

- State management for storing retrieved posts, individual post details, and form fields.
- Asynchronous data fetching using the browser's native fetch() method for all HTTP methods (GET, POST, PUT/PATCH, DELETE).
- Form validation and real-time input handling for post creation and updates.
- Client-side routing across page views using React Router.

Backend (Node.js with Express - ES Modules .mjs**):**

- loadEnvironment.mjs: Parses configuration variables from server/.env.
- db/conn.mjs: Instantiates MongoClient and exports the connected database instance.
- routes/posts.mjs: Express Router defining REST endpoints for blog operations using native MongoDB collection methods (find, findOne, insertOne, updateOne, deleteOne).
- index.mjs: Primary entry point running the Express server on designated ports.

Database (MongoDB Atlas):

- Hosting post data on MongoDB Atlas cloud infrastructure.
- Interfacing via the official native MongoDB Node.js driver (mongodb package) instead of an ORM/ODM.
- Database collections storing post documents containing fields such as _id, title, content, author, and creation timestamp.

Code:
GitHub Repository: https://github.com/KalpanaBhaskar/Web-Technologies-Laboratory/tree/main/week_9

Output:
A working full-stack Blog/Post Management Application featuring a React multi-page frontend interacting with an Express REST API using native fetch(), powered by MongoDB Atlas.

← All posts

My First Blog!

By **Kalpana Bhaskar** October 4, 2026 1 min read technology AI

Hey all! Hope you have a great time reading these tech blogs. Stay tuned for more.

Edit Post Delete Post

Discussion · 0 comments

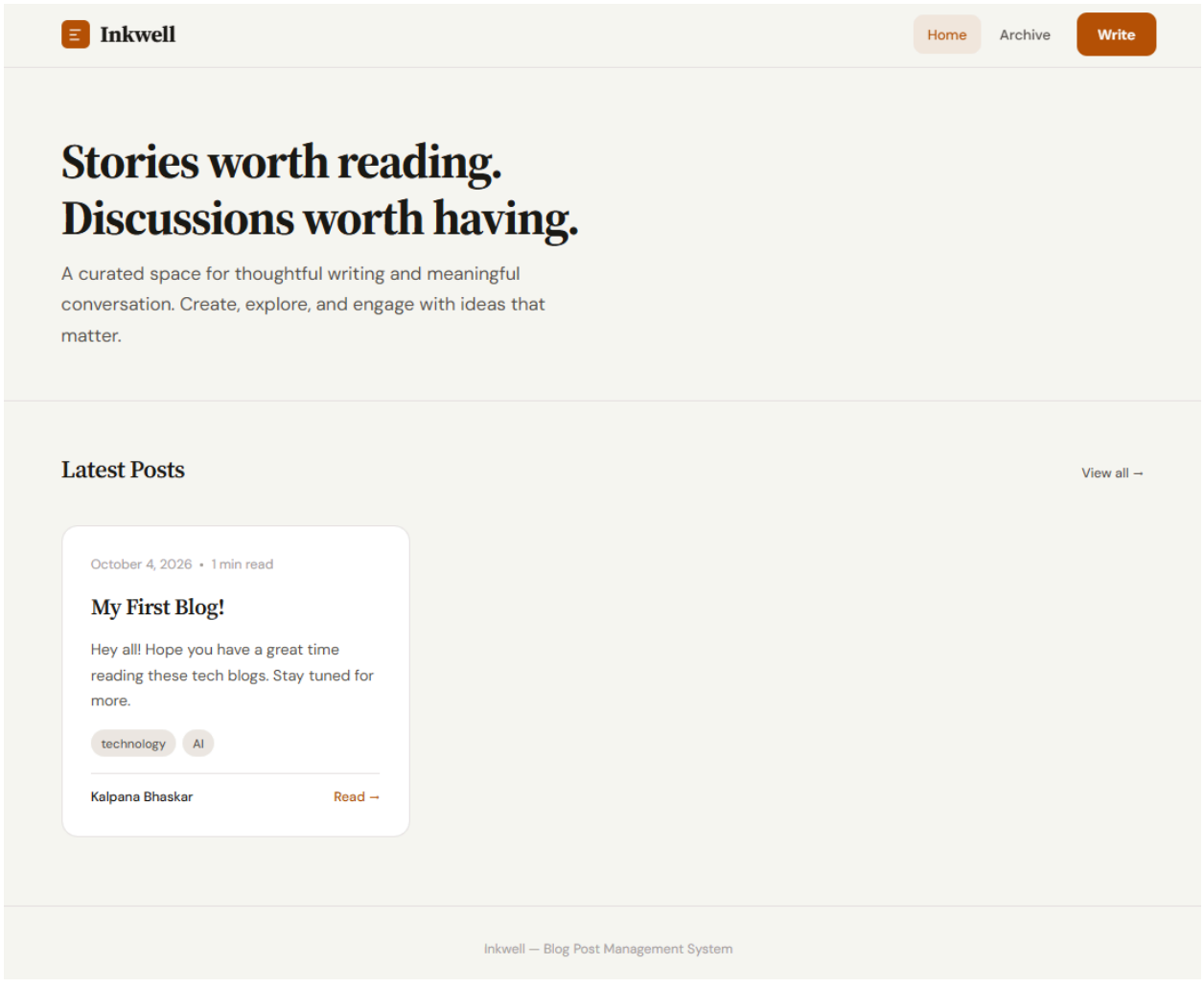
Name

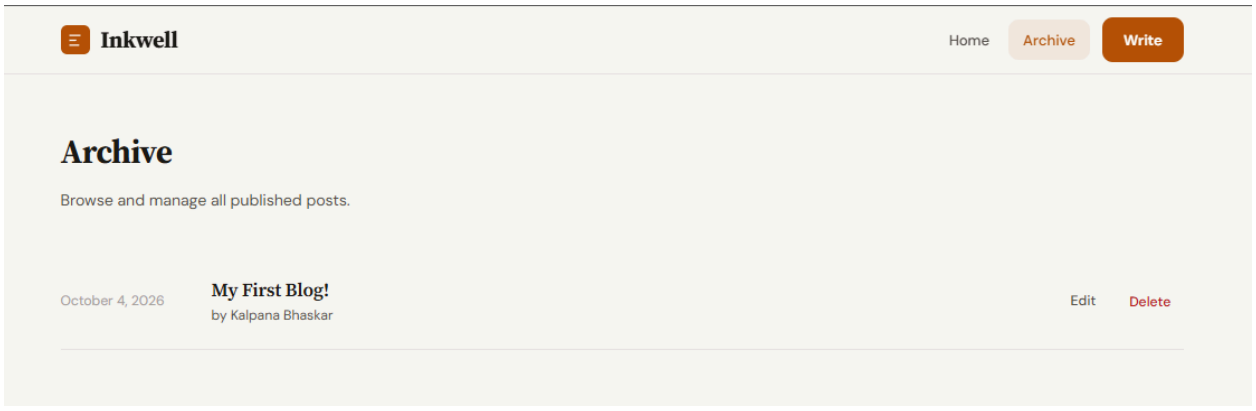
Your name

Comment

Share your thoughts...

Post Comment


Inkwell
Home
Archive
Write
Stories worth reading.
Discussions worth having.
A curated space for thoughtful writing and meaningful conversation. Create, explore, and engage with ideas that matter.
Latest Posts
View all →
October 4, 2026 • 1 min read
My First Blog!
Hey all! Hope you have a great time reading these tech blogs. Stay tuned for more.
technology
AI
Kalpana Bhaskar
Read →
Inkwell — Blog Post Management System


Inkwell
Home
Archive
Write
Archive
Browse and manage all published posts.
October 4, 2026
My First Blog!
by Kalpana Bhaskar
Edit
Delete

Inkwell

Home Archive Write

Write a New Post

Share your ideas with the Inkwell community.

Title

An engaging title for your post

Author

Your name

Content

Write your story here. Separate paragraphs with blank lines.

Tip: Use blank lines to separate paragraphs.

Excerpt (optional)

A brief summary shown in post listings

Tags (comma-separated)

technology, design, life

Cancel Publish Post